



School of Technology

A

Project Report

On

**ORBITSENSE: A MACHINE LEARNING FRAMEWORK FOR
PREDICTIVE SPACE DEBRIS COLLISION RISK
ASSESSMENT**

Submitted in partial fulfillment of the requirements for the degree of

B. Tech. in Computer Science Engineering

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CERTIFICATE

This is to certify that the project report entitled "*OrbitSense: A Machine Learning Framework for Predictive Space Debris Collision Risk Assessment*" submitted by *Shrestha Mal* (25WU0102258), *Saanvi Eluri* (25WU0101114), *Remala Bhavya Sree* (25WU0101108), *Sushennii Thangellapally* (25WU0101143) and *Syed Mohammed Awais Khadiri* (25WU0102283) in partial fulfillment of the requirements for the award of the degree of B.Tech. at Woxsen University is a bonafide record of work carried out by the students under my supervision and guidance.

The work embodied in this project report has been carried out by the candidates and has not been submitted elsewhere for a degree.

Signature of Mentor

Dr. Dharavat Nagaraju

Date: 07-10-2026

DECLARATION OF THE CANDIDATES

We hereby declare that the project work entitled "*OrbitSense: A Machine Learning Framework for Predictive Space Debris Collision Risk Assessment*" submitted to the School of Technology, Woxsen University, in partial fulfillment of the requirements for the award of the degree of B.Tech., is our original work and has been carried out under the guidance of *Dr. Dharavat Nagaraju*.

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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ABSTRACT

Low Earth Orbit (LEO) is increasingly crowded. More than 40,000 objects are tracked, of which only about one third are operational satellites, and an estimated 1.2 million fragments between 1 and 10 cm remain untracked. Collision-risk intelligence is currently provided mostly by commercial and government platforms built for large operators, leaving academic missions, CubeSat teams and students without an accessible way to explore conjunction risk.

This project presents OrbitSense, an open-data framework for conjunction-risk assessment. Public Two-Line Element (TLE) sets from CelesTrak and Space-Track are propagated with the SGP4 model for more than 16,000 objects. A three-stage screening process (bucket, coarse, fine) finds close approaches and extracts physics-based features: miss distance, relative velocity and time to closest approach. An Isolation Forest, an unsupervised anomaly-detection model from scikit-learn, ranks conjunctions by relative risk without relying on fabricated training labels. A data-driven uncertainty model, $\sigma(t) = 49.3 \sqrt{t}$ metres, fitted to multi-epoch TLE measurements, is used to compute the probability of collision (P_c) for each flagged pair. A Collision Avoidance Maneuver (CAM) simulator based on the Clohessy-Wiltshire equations estimates the effect of thruster burns. Everything is presented in an interactive live 3D dashboard built with Plotly and Dash.

The framework was validated against the 10 February 2009 collision of Iridium 33 and Cosmos 2251. OrbitSense computed a miss distance of 699.7 m against 698.0 m reported by the independent AGI STK software, an agreement of 1.7 m, and a relative velocity of 11.647 km/s against the documented value of about 11.7 km/s. The CAM simulator showed that an along-track burn of 0.5 m/s applied 120 minutes before the encounter would have increased the miss distance to about 8.8 km and reduced the probability of collision from the order of 10^{-5} to about 10^{-22} .

The work demonstrates that a physics-first, ML-assisted conjunction-risk tool can be built entirely from open data and used for education and small-mission analysis.

Keywords: Space Debris, Conjunction Analysis, SGP4, Two-Line Elements, Isolation Forest, Probability of Collision, Collision Avoidance Maneuver, Plotly Dash

TABLE OF CONTENTS

1. INTRODUCTION

- 1.1 Background
- 1.2 Motivation
- 1.3 Problem Statement
- 1.4 Objectives
- 1.5 Scope and Limitations
- 1.6 Organization of the Report
- 1.7 Significance and Applications
- 1.8 Terminology

2. LITERATURE REVIEW / TECHNOLOGY REVIEW

- 2.1 The Orbital Debris Environment
- 2.2 Public Orbital Data
- 2.3 SGP4 Propagation
- 2.4 Collision Probability
- 2.5 Anomaly Detection for Risk Ranking
- 2.6 Maneuver Modelling
- 2.7 Existing Solutions
- 2.8 Research Gap
- 2.9 Summary of the Reviewed Work
- 2.10 Design Implications

3. METHODOLOGY

- 3.1 System Overview
- 3.2 Data Collection
- 3.3 Orbital Propagation
- 3.4 Conjunction Screening and Feature Extraction
- 3.5 ML-Based Risk Ranking
- 3.6 Uncertainty Model and Probability of Collision
- 3.7 Collision Avoidance Maneuver Simulator
- 3.8 Interactive Dashboard
- 3.9 Technology Stack
- 3.10 Team Roles and Timeline
- 3.11 Why a Screening Funnel Is Needed
- 3.12 Isolation Forest in Detail
- 3.13 Probability of Collision Formulation
- 3.14 Dashboard Modes and Controls
- 3.15 Assumptions
- 3.16 Testing and Verification Strategy
- 3.17 Reproducibility

4. RESULTS AND DISCUSSION

- 4.1 Implementation Results
- 4.2 Historical Validation: Iridium 33 and Cosmos 2251
- 4.3 Collision Avoidance Maneuver Results
- 4.4 Engineering Issues Found and Fixed

- 4.5 Discussion
- 4.6 Sensitivity of Pc to Uncertainty
- 4.7 Sensitivity of the Maneuver Outcome
- 4.8 Comparison with Existing Approaches
- 4.9 Challenges Encountered
- 4.10 Individual Contributions

5. CONCLUSION AND FUTURE WORK

- 5.1 Summary of Contributions
- 5.2 Limitations
- 5.3 Future Work
- 5.4 Concluding Remarks
- 5.5 Lessons Learned
- 5.6 Societal and Educational Impact

REFERENCES

APPENDIX - I

APPENDIX - II

APPENDIX - III

Glossary of Abbreviations

LIST OF TABLES

Table 1.1: Scale of the orbital debris problem

Table 2.1: Comparison of existing solutions and OrbitSense

Table 2.2: Summary of reviewed literature and its use in this project

Table 3.1: Technology stack

Table 3.2: Team roles

Table 3.3: Project timeline

Table 3.4: Dashboard controls

Table 4.1: Historical validation of the Iridium 33 / Cosmos 2251 conjunction

Table 4.2: Probability of collision for assumed position uncertainties

Table 4.3: CAM simulator results (along-track burn, 120 min before encounter)

Table 4.4: Bugs identified and fixed

Table 4.5: Live dashboard snapshot

Table 4.6: Qualitative comparison of OrbitSense with existing approaches

Table 4.7: Individual contributions

LIST OF FIGURES

Figure 3.1: OrbitSense system architecture

Figure 3.2: Three-stage conjunction screening

Figure 4.1: Live 3D tracker

Figure 4.2: Satellite search, camera lock and telemetry

Figure 4.3: Interactive Physics Sandbox

Figure 4.4: Historical validation output

Figure 4.5: CAM simulator output

1. INTRODUCTION

Since the launch of Sputnik in 1957, decades of launches, collisions and fragmentation events have filled Earth orbit with debris. Objects in LEO travel at roughly 7 to 8 km/s, so even a centimetre-sized fragment carries enough kinetic energy to disable a spacecraft. Tracking this population and predicting close approaches, known as conjunctions, is a core part of space situational awareness.

1.1 Background

Public catalogues such as those maintained by Space-Track.org and republished by CelesTrak describe each tracked object with a Two-Line Element set (TLE). Combined with the SGP4 analytical propagator, a TLE can be used to predict an object's position and velocity at any time near its epoch. By propagating pairs of objects and finding the time of closest approach, one can estimate the miss distance and relative velocity of a conjunction, and from these assess its risk.

Operational conjunction assessment is carried out by government agencies and by commercial providers such as LeoLabs, using proprietary radar data and tools designed for large satellite operators. The authoritative studies from the NASA Orbital Debris Program Office and the ESA Space Debris Office are research publications rather than interactive tools.

1.2 Motivation

- Growing congestion in LEO and the risk of cascading collisions.
- Lack of open, interactive tools for students, CubeSat teams and academic missions.
- An opportunity to combine orbital mechanics with applied machine learning in a transparent way.
- A real, documented collision (Iridium 33 / Cosmos 2251) against which a tool can be honestly validated.

1.3 Problem Statement

Collision-risk intelligence is available mainly through enterprise-grade platforms. There is no accessible, open-data, transparent framework that lets non-specialists explore conjunction risk, understand how it is computed, and test avoidance options. This project addresses that gap.

Quantity	Approximate value
Tracked objects in orbit	More than 40,000
Share that are active satellites	About one third
Untracked fragments of 1-10 cm	About 1.2 million
Objects tracked live in OrbitSense	More than 16,000

Table 1.1: Scale of the orbital debris problem

1.4 Objectives

1. Build a pipeline to retrieve and parse public orbital data (TLE sets) from CelesTrak and Space-Track.
2. Implement SGP4-based propagation and conjunction analysis to generate physics-based risk features: miss distance, relative velocity and time to closest approach.
3. Train an ML model to classify and rank conjunction risk using these features, without replacing the underlying physics.
4. Design an interactive dashboard that flags high-risk conjunctions in a form a non-specialist can follow.
5. Validate the framework against at least one documented historical close-approach event.

1.5 Scope and Limitations

The scope covers conjunction screening and risk ranking for publicly catalogued objects using TLE data. TLE data is less accurate than the radar-derived data used by operators, so OrbitSense is an educational and analytical tool and not an operational collision-warning service. Further limitations are discussed in Chapter 5.

1.6 Organization of the Report

Chapter 2 reviews related work. Chapter 3 describes the methodology, architecture and tools. Chapter 4 presents implementation results, validation and discussion. Chapter 5 concludes and outlines future work.

1.7 Significance and Applications

Although OrbitSense is a student project, the problem it addresses is practical. Small satellite teams, university CubeSat programmes and space-education initiatives increasingly launch hardware into LEO, yet they rarely have access to commercial conjunction-assessment services. An open framework allows such groups to understand how a close approach is detected, how its probability is estimated, and what a maneuver would change.

- Education: students can see orbital mechanics, uncertainty and decision-making in one interactive environment.
- Small missions: teams can screen their own spacecraft against the public catalogue and decide whether to seek professional support.
- Research: the open pipeline can serve as a baseline for testing alternative uncertainty models and ranking methods.
- Awareness: a live visualization of more than 16,000 objects communicates the scale of orbital congestion to a general audience.

1.8 Terminology

The following terms are used throughout this report. A conjunction is a predicted close approach between two orbiting objects. The miss distance is the separation between the two objects at the time of closest approach (TCA). The relative velocity is the speed of one object with respect to the other at TCA. The probability of collision (P_c) is the likelihood that the two objects actually collide, given their position uncertainties and sizes. A collision avoidance maneuver (CAM) is a thruster burn performed to change an orbit and increase the miss distance.

2. LITERATURE REVIEW / TECHNOLOGY REVIEW

2.1 The Orbital Debris Environment

The NASA Orbital Debris Program Office and the ESA Space Debris Office publish regular assessments of the tracked and modelled debris population [1], [2]. Kessler and Cour-Palais showed that collisions between objects can generate enough fragments to produce a self-sustaining cascade, now called the Kessler syndrome [11]. The 2009 collision between Iridium

33 and Cosmos 2251 is the first accidental hypervelocity collision between two intact satellites and created thousands of tracked fragments [9].

2.2 Public Orbital Data

Space-Track.org, operated by the 18th Space Defense Squadron of the U.S. Space Force, distributes the general-perturbations catalogue as TLEs [4]. CelesTrak republishes this data in convenient formats and also provides SOCRATES, a service that lists predicted close approaches [3]. In this project, SOCRATES predictions for the 2009 event are used only as context for the choice of uncertainty values.

2.3 SGP4 Propagation

The SGP4 model was introduced in Spacetrack Report No. 3 [5] and later revisited and corrected by Vallado et al. [6]. It models the main perturbations on near-Earth orbits, including the Earth's oblateness (J_2) and atmospheric drag, using analytical expressions. SGP4 is the standard propagator for TLE data, and its accuracy degrades as the prediction moves away from the TLE epoch.

2.4 Collision Probability

Probability of collision (P_c) combines the miss distance, the position uncertainty of both objects and the combined hard-body radius. A common formulation integrates a two-dimensional Gaussian over the combined hard-body circle in the encounter plane [10]. P_c depends strongly on the assumed uncertainty, which is why the choice of uncertainty model matters.

2.5 Anomaly Detection for Risk Ranking

Labelled examples of real collisions are extremely rare, so supervised classification of conjunction risk would require either synthetic or fabricated labels. Isolation Forest [7] avoids this by isolating unusual observations through random partitioning; the fewer splits needed to isolate a point, the more anomalous it is. Applied to physics-derived features, it can rank which conjunctions are unusual without labels. The implementation used here is from scikit-learn [13].

2.6 Maneuver Modelling

The Clohessy-Wiltshire (Hill) equations describe the motion of one object relative to another in near-circular orbits [8]. They allow the effect of a small thruster burn on the relative position at a later time to be estimated analytically.

2.7 Existing Solutions

Solution	Nature	Limitation for students
LeoLabs / Slingshot Aerospace	Commercial, enterprise-grade	Paid; built for operators and governments [12]
NASA ODPO / ESA	Authoritative research	Publications, not an interactive tool
CelesTrak SOCRATES	Predicted conjunction lists	Lists results; no interactive risk or maneuver exploration
OrbitSense (this project)	Open data, transparent method	TLE-based accuracy; educational use

Table 2.1: Comparison of existing solutions and OrbitSense

2.8 Research Gap

Existing options are either closed and costly or authoritative but non-interactive. OrbitSense fills this gap with an open, physics-first and explainable tool that combines live tracking, ML-assisted ranking, probability of collision and maneuver simulation, and that is independently validated against a known event.

2.9 Summary of the Reviewed Work

Table 2.2 summarizes how each reviewed source contributes to the design of OrbitSense. The sources fall into three groups: those that define the problem, those that supply data and models, and those that supply methods for ranking and maneuver analysis.

Source	Contribution	Use in OrbitSense
NASA ODPO, ESA [1], [2]	Debris population statistics	Problem context and scale
Kessler and Cour-Palais [11]	Collisional cascade concept	Motivation for risk assessment

Source	Contribution	Use in OrbitSense
CelesTrak, Space-Track [3], [4]	Open TLE catalogue and SOCRATES	Data source and context for uncertainty
Hoots and Roehrich; Vallado et al. [5], [6]	SGP4 propagation	Position and velocity prediction
Chan [10]	Collision probability theory	Pc computation
Liu et al. [7]	Isolation Forest	Unsupervised risk ranking
Clohessy and Wiltshire [8]	Relative-motion equations	CAM simulator
Kelso [9]	Analysis of the 2009 collision	Validation case study

Table 2.2: Summary of reviewed literature and its use in this project

2.10 Design Implications

Three design decisions follow from the review. First, because SGP4 accuracy depends on the age of the TLE, the uncertainty used for Pc should grow with TLE age rather than be a fixed number. Second, because labelled collision data is almost nonexistent, an unsupervised ranking method is more defensible than a supervised classifier trained on synthetic labels. Third, because the 2009 collision is thoroughly documented and has been analysed with independent software, it is a suitable validation case.

3. METHODOLOGY

3.1 System Overview

OrbitSense follows a physics-first design: orbital mechanics generates the risk features, and machine learning ranks the risk those features represent. Machine learning never substitutes for propagation or conjunction analysis.



Figure 3.1: OrbitSense system architecture

3.2 Data Collection

TLE sets for the active satellite catalogue are retrieved from CelesTrak, with Space-Track as a second public source. During development a fetch silently returned zero satellites. The pipeline was therefore changed to validate every response, cache the last good download locally, and fail with an explicit error instead of hiding the problem. At the time of the demonstration the system tracked 16,060 objects.

3.3 Orbital Propagation

Each TLE is propagated using the SGP4 implementation in the sgp4 and Skyfield libraries to obtain position and velocity vectors in an Earth-centred frame at the required times. Positions are refreshed periodically (default 30 seconds) for live visualization.

3.4 Conjunction Screening and Feature Extraction

Testing all pairs among 16,000 objects over a time window is computationally impractical, so a three-stage funnel is used.

- **Bucket stage:** objects are grouped by spatial cells so that only objects in neighbouring cells are compared.
- **Coarse stage:** candidate pairs are checked at coarse time steps against a generous distance threshold.
- **Fine stage:** surviving pairs are refined around the time of closest approach to find the exact miss distance, relative velocity and time to closest approach (TCA).



Figure 3.2: Three-stage conjunction screening

3.5 ML-Based Risk Ranking

The features miss distance, relative velocity and time to closest approach for each screened conjunction are standardized and passed to an Isolation Forest. Conjunctions with the highest anomaly scores are ranked as the highest relative risk and highlighted in the dashboard. Because there are no ground-truth labels for real collisions at this scale, an unsupervised model was chosen deliberately, and no synthetic labels were fabricated for training.

3.6 Uncertainty Model and Probability of Collision

To avoid using an arbitrary assumed uncertainty, position error growth was estimated from real multi-epoch TLE measurements. The fitted relationship is $\sigma(t) = 49.3 \sqrt{t}$ metres, where t is the age of the TLE. For every flagged pair, the uncertainty is computed from each satellite's actual current TLE age, and P_c is evaluated by integrating a two-dimensional Gaussian over the combined hard-body circle in the encounter plane. A combined hard-body radius of 10 m was used for the validation case.

3.7 Collision Avoidance Maneuver Simulator

The CAM simulator applies an along-track velocity change (Δv) at a chosen time before the encounter and propagates the relative motion using the Clohessy-Wiltshire equations. It then recomputes the miss distance and P_c . An early version using Kepler re-propagation diverged from SGP4 by about 50 km because it ignored J2 perturbation, which motivated the Clohessy-Wiltshire formulation.

3.8 Interactive Dashboard

The dashboard is built with Plotly and Dash. It shows a textured 3D Earth with all tracked objects, red and orange highlights for high- and moderate-risk conjunctions, a heads-up display with live statistics, a satellite search box with camera lock, and a telemetry card. A Physics Sandbox provides sliders for Δv (plus or minus 5 m/s) and tracking uncertainty σ (100 to 2000 m) that recompute miss distance and P_c instantly. Student and Advanced modes adjust the level of detail. Updates use Dash's Patch mechanism so that the camera is not reset on refresh.

3.9 Technology Stack

Component	Technology
Language	Python
Propagation	SGP4, Skyfield
Data processing	Pandas, NumPy
Machine learning	scikit-learn (Isolation Forest)
Dashboard	Plotly, Dash
Data sources	CelesTrak, Space-Track APIs
Version control	Git and GitHub

Table 3.1: Technology stack

3.10 Team Roles and Timeline

Member	Role
Syed Mohammed Awais Khadiri	Project Manager and Research Lead
Saanvi Eluri	Data Acquisition and Orbital Mechanics
Shrestha Mal	Machine Learning and Risk Analysis
Sushennii Thangellapally	Visualization and Dashboard Developer
Remala Bhavya Sree	Testing, Validation and Deployment

Table 3.2: Team roles

Weeks	Activity
1-4	Setup, live pipeline, SGP4 propagation, Initial Proposal
5-8	ML risk ranking, validation, uncertainty model, live Pc, CAM simulator, Mid-Term review
9-12	Search and camera lock, Physics Sandbox, final polish, full report, final review

Table 3.3: Project timeline

3.11 Why a Screening Funnel Is Needed

A catalogue of N objects contains $N(N-1)/2$ distinct pairs. For $N = 16,000$ this is about 128 million pairs, and each pair would need to be checked at many time steps. A direct approach is therefore too slow for an interactive tool. The funnel reduces the work by discarding pairs that cannot possibly be close. The bucket stage uses a cheap spatial test, the coarse stage uses a generous distance threshold at coarse time steps, and only the small number of surviving pairs reach the precise but expensive fine stage. The funnel is designed so that the early stages are conservative: they may keep pairs that later turn out to be safe, but they should not discard pairs that could be close.

3.12 Isolation Forest in Detail

An Isolation Forest builds many random binary trees. At each node a random feature and a random split value are chosen, so that the data is recursively partitioned. Points that are unusual need fewer splits to be isolated, so their average path length across the trees is short. The anomaly score of a point is derived from this average path length, normalized by the expected path length for a dataset of the same size, so that scores close to 1 indicate strong anomalies and scores well below 0.5 indicate normal points [7].

For OrbitSense this means that a conjunction with a very small miss distance, a high relative velocity and a near-term TCA will tend to be isolated quickly and rank highly. The model does not know that such a conjunction is dangerous. It only knows that it is rare compared with the other screened pairs, which is why the ranking is presented together with the physically computed P_c .

3.13 Probability of Collision Formulation

In the encounter plane, the relative position of the two objects at TCA is described by the miss vector. The position error of both objects is modelled as a zero-mean Gaussian with combined standard deviation σ . P_c is the integral of this Gaussian density over a circle whose radius equals the combined hard-body radius of the two objects and which is centred on the miss vector [10]. When the miss distance is small compared with σ , P_c is governed mainly by the ratio of the hard-body area to the uncertainty area. When the miss distance is much larger than σ , the exponential tail dominates and P_c falls very rapidly, which explains values as small as 10^{-92} for a routine 20 km conjunction.

3.14 Dashboard Modes and Controls

Control	Range / options	Effect
Mode toggle	Student / Advanced	Switches between plain-language risk text and numerical detail
Delta-v slider	Plus or minus 5 m/s	Recomputes miss distance and Pc for the selected conjunction
Sigma slider	100 to 2000 m	Shows how tracking precision changes risk
Search and lock	Name or NORAD ID	Centres and follows a chosen satellite
Refresh interval	Selectable, default 30 s	Controls how often positions are updated
Pause live refresh	On / off	Freezes the view for inspection

Table 3.4: Dashboard controls

3.15 Assumptions

- TLE data from CelesTrak and Space-Track is accurate enough to illustrate conjunction analysis, although it is not operator-grade.
- Position error grows with TLE age according to the fitted relationship $\sigma(t) = 49.3 \sqrt{t}$ metres.
- The combined hard-body radius is a constant 10 m for the validation case.
- Relative motion over the maneuver interval is adequately described by the Clohessy-Wiltshire equations.
- The screened pairs form a population in which unusual geometry is a useful proxy for relative risk.

3.16 Testing and Verification Strategy

Testing was carried out at four levels. Module-level checks confirmed that data fetching rejects empty or malformed responses and that the cache is used when the network source fails. Numerical checks compared propagated positions against the SGP4 reference output. System-level checks confirmed that the dashboard refreshes without resetting the camera. Finally, the historical validation compared the complete pipeline against an independent result from AGI

STK. This layered approach meant that errors such as the dropped fractional second were traced to a specific module instead of being discovered only in the final output.

3.17 Reproducibility

All code is version-controlled in a GitHub repository [repository link]. The repository contains the main dashboard script, the historical validation script and the CAM simulator script, together with a list of required Python packages. Downloaded TLE data is cached locally so that results can be reproduced without repeated downloads, and the historical validation uses a fixed event time so that its output is deterministic.

4. RESULTS AND DISCUSSION

4.1 Implementation Results

All planned modules have been implemented and run end to end: live data ingestion, SGP4 propagation for more than 16,000 objects, scalable screening, Isolation Forest ranking, empirical-uncertainty P_c , the CAM simulator and the interactive dashboard with the Physics Sandbox.

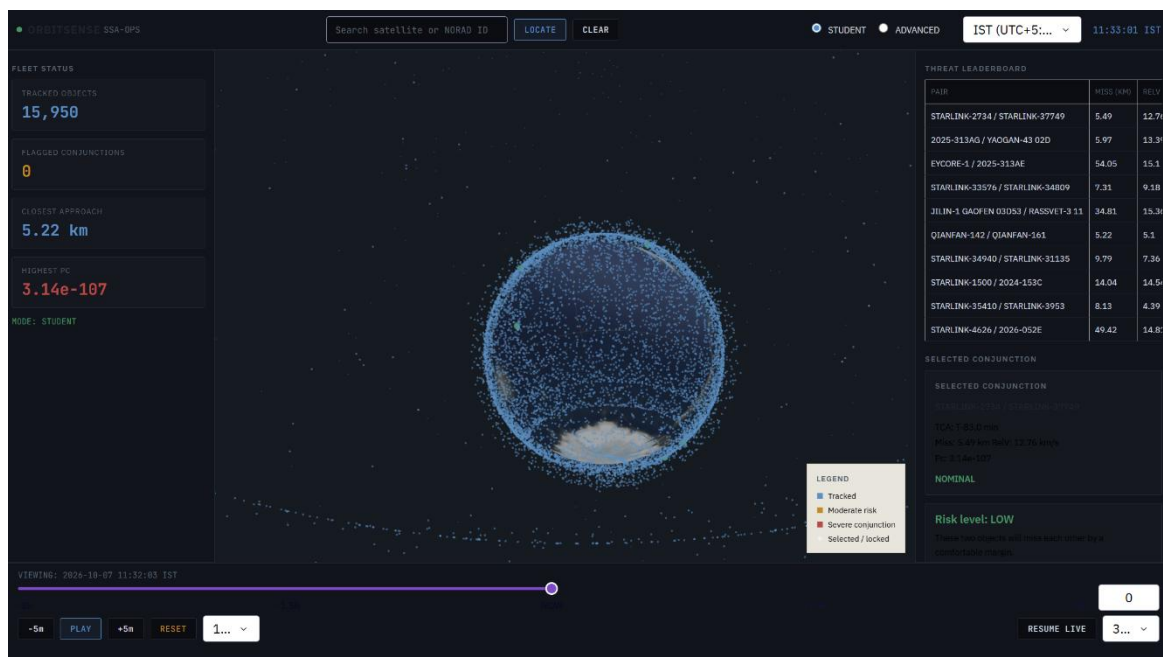


Figure 4.1: Live 3D tracker

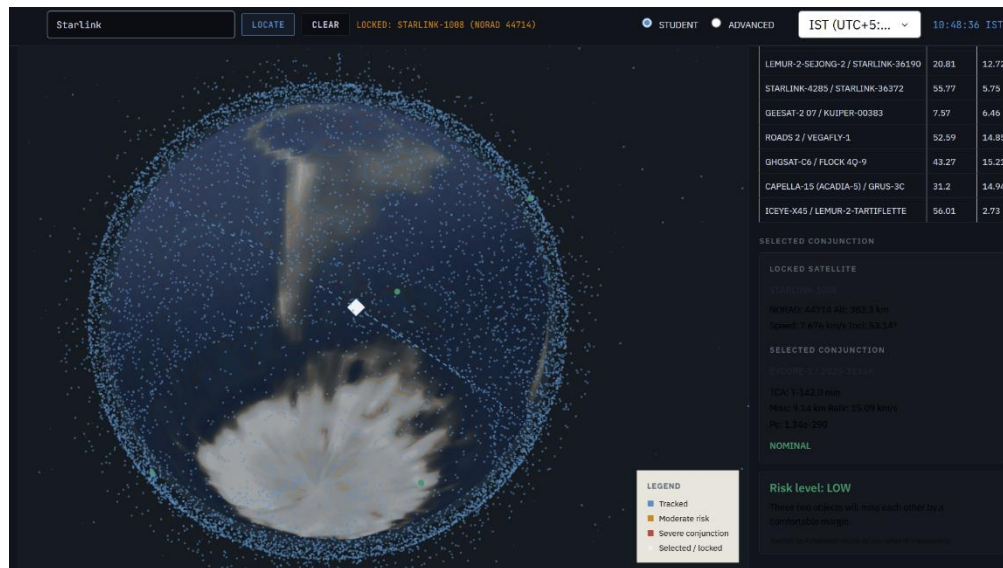


Figure 4.2: Satellite search, camera lock and telemetry

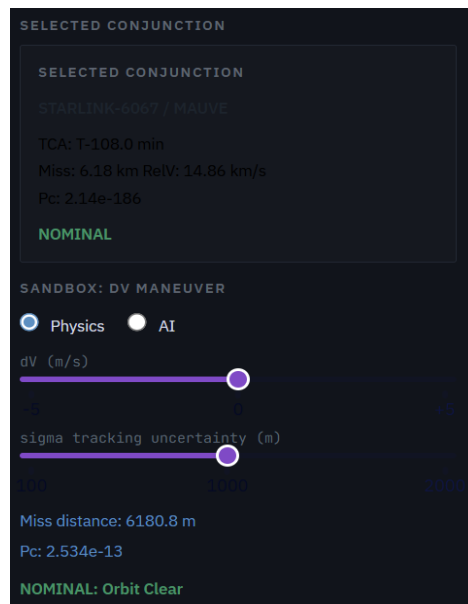


Figure 4.3: Interactive Physics Sandbox

Metric	Value
Tracked objects	16,060
Flagged conjunctions	10
Closest approach	3.84 km
Highest Pc among flagged pairs	7.18e-51
Refresh interval	30 s (selectable)

Table 4.5: Live dashboard snapshot

The snapshot shows that the highest P_c among the flagged pairs at that moment was extremely small. This is expected: most flagged pairs are unusual compared with the population but are not dangerous. The risk ranking highlights which conjunctions deserve attention, and P_c quantifies how serious they are.

4.2 Historical Validation: Iridium 33 and Cosmos 2251

On 10 February 2009 the Iridium 33 and Cosmos 2251 satellites collided at about 790 km altitude. OrbitSense propagated TLEs for both objects and located the closest approach at the documented time.

Quantity	OrbitSense	Reference
Miss distance	699.7 m	698.0 m (AGI STK)
Agreement	1.7 m	-
Relative velocity	11.647 km/s	About 11.7 km/s (documented)

Table 4.1: Historical validation of the Iridium 33 / Cosmos 2251 conjunction

```

=====
HISTORICAL VALIDATION: Iridium 33 / Cosmos 2251, 10 Feb 2009
=====
Closest approach found at documented time -0.0s
  OrbitSense computed miss distance : 699.7 m
  Independently verified (AGI STK)  : 698.0 m (agreement: 1.7 m)
  OrbitSense computed relative velocity : 11.647 km/s
  Documented relative velocity          : ~11.7 km/s

Probability of Collision (Pc), combined hard-body radius = 10 m:
(sigma = assumed combined position uncertainty; SOCRATES predictions for
this exact event ranged 117 m-1.8 km across the week before, per CelesTrak)
  sigma= 300 m -> Pc = 3.66e-05
  sigma= 500 m -> Pc = 7.51e-05
  sigma= 700 m -> Pc = 6.19e-05
  sigma= 1000 m -> Pc = 3.91e-05
  sigma= 1500 m -> Pc = 1.99e-05
  sigma= 1800 m -> Pc = 1.43e-05

For comparison, a routine 20 km conjunction -> Pc = 6.95e-92
=====

```

Figure 4.4: Historical validation output

The miss distance agrees with the independent AGI STK result to within 1.7 m, and the relative velocity matches the documented value. This confirms that the propagation and closest-approach logic reproduce a professional result for a real event.

P_c was evaluated for several assumed combined position uncertainties. SOCRATES predictions for this event in the week before ranged from 117 m to 1.8 km in miss distance, so a range of sigma values is reported rather than a single number.

Sigma (m)	P_c
300	3.66e-05
500	7.51e-05
700	6.19e-05
1000	3.91e-05
1500	1.99e-05
1800	1.43e-05
Routine 20 km conjunction (for comparison)	6.95e-92

Table 4.2: Probability of collision for assumed position uncertainties

For the 2009 event, P_c stays in the range of about 10^{-5} across the plausible uncertainty range, many orders of magnitude above a routine 20 km conjunction. P_c is not monotonic in sigma: it peaks near the miss distance and falls for larger and smaller sigma, as expected for a Gaussian model.

4.3 Collision Avoidance Maneuver Results

Delta-v (mm/s)	Miss distance (m)	P_c
0 (no burn)	699.7	3.914e-05
10	665.0	4.008e-05
30	736.6	3.812e-05
50	950.4	3.183e-05
100	1715.7	1.147e-05
300	5232.0	5.689e-11
500	8822.2	6.289e-22
1000	17828.3	4.792e-74

Table 4.3: CAM simulator results (along-track burn, 120 min before encounter)

```

Along-track burn, 120 min before encounter:
dv=   0 mm/s -> miss=   699.7 m   Pc=3.914e-05
dv=  10 mm/s -> miss=   665.0 m   Pc=4.008e-05
dv=  30 mm/s -> miss=   736.6 m   Pc=3.812e-05
dv=  50 mm/s -> miss=   950.4 m   Pc=3.183e-05
dv= 100 mm/s -> miss=  1715.7 m   Pc=1.147e-05
dv= 300 mm/s -> miss=  5232.0 m   Pc=5.689e-11
dv= 500 mm/s -> miss=  8822.2 m   Pc=6.289e-22
dv=1000 mm/s -> miss= 17828.3 m   Pc=4.792e-74

```

Figure 4.5: CAM simulator output

A burn of 0.5 m/s applied two hours before the encounter increases the miss distance to about 8.8 km and reduces P_c from about $3.9e-05$ to about $6.3e-22$. Very small burns of 10 mm/s slightly worsen the outcome (P_c rises to $4.0e-05$), showing that a maneuver must be large enough to move the objects out of the high-probability region; a tiny burn is not automatically safe.

4.4 Engineering Issues Found and Fixed

Issue	Cause and fix
CelesTrak returned 0 satellites silently	Added response validation and local caching; failures are now explicit
CAM miss distance was 9.3 km off	A dropped fractional second; a 0.8 s timing error corresponds to about 9.3 km at this velocity
Kepler re-propagation diverged from SGP4 by about 50 km	Unmodelled J2 perturbation; replaced with Clohessy-Wiltshire equations
Dashboard camera reset every 30 s	Rebuilt the refresh around Dash Patch(); no layout or camera operations are sent on refresh

Table 4.4: Bugs identified and fixed

4.5 Discussion

First, the physics-first design worked: a well-implemented SGP4 and closest-approach search is sufficient to reproduce a professional result, and ML is used only for ranking. Second, an unsupervised model is an honest choice when labelled collision data does not exist, although it measures how unusual a conjunction is, not how dangerous it is; this is why P_c is reported alongside the ranking. Third, P_c is highly sensitive to the assumed uncertainty, which is why

the project derives $\sigma(t)$ from data and exposes σ as a slider. Fourth, the debugging record shows that small modelling errors such as timing or J2 effects lead to large physical errors, so validation against an independent reference is essential.

[Optional: add any quantitative comparison between Isolation Forest ranking and a physics-only ranking, such as ranking agreement or the number of top-ranked pairs shared, if you have measured it.]

4.6 Sensitivity of P_c to Uncertainty

The results in Table 4.2 show that, for the 2009 event, P_c varies by a factor of about five across the tested uncertainty range, from $1.43\text{e-}05$ at 1800 m to $7.51\text{e-}05$ at 500 m. The variation is far smaller than the difference between this event and a routine conjunction, but it is not negligible when decisions are made near a threshold. The maximum occurs when σ is of the same order as the miss distance, which is a well-known property of this probability model. It also explains why the Physics Sandbox exposes σ as a slider: users can see for themselves that P_c is not a single fixed number but depends on how well the objects are tracked.

4.7 Sensitivity of the Maneuver Outcome

Table 4.3 shows three regimes. For burns up to about 30 mm/s the miss distance changes only slightly and P_c stays near $4\text{e-}05$, so the maneuver is effectively useless. Between 50 and 100 mm/s the miss distance rises to 1.7 km and P_c drops by about a factor of three. From 300 mm/s upward the miss distance exceeds 5 km and P_c falls by many orders of magnitude. The practical lesson is that avoidance requires moving the objects well beyond the uncertainty envelope, not merely changing their separation slightly. The 0.5 m/s burn is a small velocity change for a satellite, yet applied two hours early it moves the encounter far outside the region of significant probability.

4.8 Comparison with Existing Approaches

Feature	Enterprise platforms	Agency publications	OrbitSense
Cost	Paid	Free to read	Free
Data	Proprietary radar and optical	Aggregated studies	Public TLEs
Interactivity	Operator dashboards	None	Live 3D dashboard

Feature	Enterprise platforms	Agency publications	OrbitSense
Transparency of method	Limited	High	High (open code)
Maneuver exploration	Yes, for customers	No	Yes, simulated
Intended users	Large operators, defence	Researchers	Students, CubeSat teams

Table 4.6: Qualitative comparison of OrbitSense with existing approaches

The comparison is qualitative. OrbitSense does not claim to match the accuracy of enterprise platforms, which use better data. Its contribution is accessibility and transparency, together with a validated core calculation.

4.9 Challenges Encountered

The main challenge was scale. Running conjunction screening across the full catalogue required the funnel design described in Section 3.11, because a naive pairwise approach was not feasible. The second challenge was correctness: several of the bugs in Table 4.4 produced results that looked plausible but were physically wrong, and they were found only by comparing against independent references. The third challenge was interface behaviour, since a live dashboard that resets the camera every 30 seconds is unusable. Fixing this required changing how updates are sent to the browser rather than adjusting any single parameter.

4.10 Individual Contributions

Member	Contribution to the project
Syed Mohammed Awais Khadiri	Project management, research direction, report coordination
Saanvi Eluri	Data acquisition, TLE handling, orbital mechanics and propagation
Shrestha Mal	Isolation Forest ranking, risk analysis, feature extraction
Sushennii Thangellapally	3D visualization, dashboard, Physics Sandbox
Remala Bhavya Sree	Testing, historical validation, deployment

Table 4.7: Individual contributions

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

6. An open-data pipeline that retrieves and validates public TLE data for more than 16,000 objects.
7. A scalable bucket, coarse and fine conjunction-screening method built on SGP4 propagation.
8. An Isolation Forest risk-ranking model that uses physics-derived features and no fabricated labels.
9. A data-driven uncertainty model, $\sigma(t) = 49.3 \sqrt{t}$ m, and live probability-of-collision for flagged pairs.
10. A Clohessy-Wiltshire CAM simulator and an interactive Physics Sandbox.
11. Independent validation against the Iridium 33 / Cosmos 2251 collision, agreeing with AGI STK to within 1.7 m.

5.2 Limitations

- TLE data is less accurate than operator-grade radar or ephemeris data, so results are for education and analysis, not operational warnings.
- Validation is based on one historical event; more events would strengthen confidence.
- Isolation Forest ranks how unusual a conjunction is and cannot be evaluated with precision and recall without labelled data.
- The Pc model assumes Gaussian position errors and a spherical combined hard-body radius.
- The Clohessy-Wiltshire model assumes near-circular orbits and small separations.
- Objects below the tracking threshold are not represented.

5.3 Future Work

- Interactive time-scrubber and orbit playback.
- 2D ground-track map alongside the 3D globe.
- Altitude-shell (LEO, MEO, GEO) filtering.

- Validation on additional documented close approaches and comparison with SOCRATES.
- Collaboration with an academic CubeSat mission for live testing.
- Open-source release and a possible publication.

5.4 Concluding Remarks

OrbitSense shows that a transparent, physics-first and ML-assisted conjunction-risk framework can be built entirely from open data and validated against a real event. It offers students and small satellite teams a practical way to explore how conjunction risk is computed and how a maneuver can avoid it.

5.5 Lessons Learned

- Validate early against an independent reference; plausible-looking output is not evidence of correctness.
- Make failures loud: the silent empty download shows why data pipelines should reject bad input explicitly.
- Choose methods that match the data available; an unsupervised model is more honest than one trained on invented labels.
- Expose uncertainty to the user instead of hiding it behind a single number.
- Divide work by clear roles but review each other's modules, since many bugs sit at the boundary between modules.

5.6 Societal and Educational Impact

Greater awareness of orbital debris supports responsible use of space. By making conjunction analysis open and understandable, OrbitSense can help train the next generation of engineers who will operate and regulate satellites. It can also help small missions make informed decisions about whether their spacecraft are at risk.

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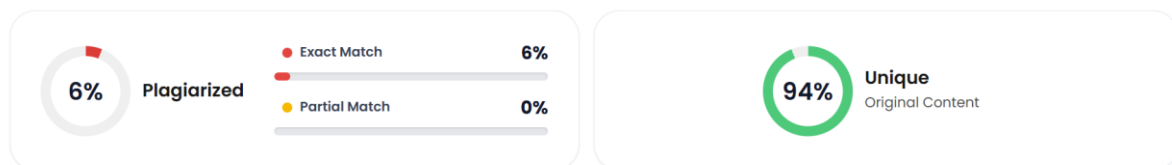
APPENDIX - I

- **GitHub:** <https://github.com/Shrestha3511/orbitsense>

Shrestha3511	Merge pull request #6 from Shrestha3511/archive-first-test-document...	44:9062 · 6 hours ago	21 Commits
github/workflows	feat: harden canonical app, add tests, docs, and CI	3 days ago	
archive	Move minimal TLE check into archive	6 hours ago	
assets	Add dashboard screenshot to README	2 days ago	
tests	Fix tracker search lock flow and preserve static traces on focus	3 days ago	
.gitattributes	Initial commit	3 days ago	
.gitignore	feat: harden canonical app, add tests, docs, and CI	3 days ago	
LICENSE	feat: harden canonical app, add tests, docs, and CI	3 days ago	
PRIVACY.md	Fix tracker search lock flow and preserve static traces on focus	3 days ago	
README.md	Document HTML, demos and archived connectivity check	6 hours ago	
TERMS.md	Fix tracker search lock flow and preserve static traces on focus	3 days ago	
cam_simulator.py	Initial commit	3 days ago	
historical_validation.html	Initial commit	3 days ago	
historical_validation.py	feat: harden canonical app, add tests, docs, and CI	3 days ago	
live_tracker_v8.py	Update live_tracker v8.py	2 days ago	
orbit_view.html	Initial commit	3 days ago	
orbit_visual.py	feat: harden canonical app, add tests, docs, and CI	3 days ago	
requirements-dev.txt	feat: harden canonical app, add tests, docs, and CI	3 days ago	
requirements.txt	feat: harden canonical app, add tests, docs, and CI	3 days ago	
risk_map.html	Initial commit	3 days ago	
risk_map.py	feat: harden canonical app, add tests, docs, and CI	3 days ago	
uncertainty_model.py	Initial commit	3 days ago	

APPENDIX - II

Plagiarism report:



Mentor Review Approval Form:

6. Mentor Approval Decision

☒ Approved for Final Evaluation

☐ Conditionally Approved (corrections required before evaluation)

☐ Not Approved (re-review required)

Conditions / Reasons (if applicable):

7. Signatures & Verification

Dr. Dharmendra Nigam (Mentor Name & Designation)

Project Coordinator

Signature & Date: Dr. Nigam (Signature) 06/10/26

Signature & Date: (Signature) 06/10/26

Signature, Stamp & Date: (Signature) 06/10/26

Conceptual 1 — Mentor Review Approval Form | For Official Use Only | Submit to the Examination / Academic Office after mentor sign-off.

Page 3 of 3

APPENDIX - III

Glossary of Abbreviations

Abbreviation	Meaning
LEO	Low Earth Orbit
TLE	Two-Line Element set
SGP4	Simplified General Perturbations model 4
TCA	Time of Closest Approach
Pc	Probability of Collision
CAM	Collision Avoidance Maneuver
ODPO	Orbital Debris Program Office
ESA	European Space Agency
SOCRATES	Satellite Orbital Conjunction Reports Assessing Threatening Encounters in Space
AGI STK	Analytical Graphics Systems Tool Kit (independent orbital analysis software)
J2	Second zonal harmonic of the Earth's gravity field (oblateness)
ML	Machine Learning

Table A.1: Glossary of abbreviations